TanitAD-4B: Hierarchical Latent World Models with Typed Abstraction Interfaces for Data-Efficient Autonomous Driving

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November 2026

Abstract

End-to-end driving models have won the architecture argument, yet they remain data-hungry, opaque, unaware of their own limits and expensive at inference time. We present *TanitAD-4B*, a programme that attacks all four weaknesses with one design principle: *drive at every level of abstraction, and make every level accountable*. The 4B architecture factorises driving into four brains (operative, tactical, strategic and fallback), each with its own clock, horizon, state space and predictor. Goals condition strictly downward and latents flow upward through stop-gradient barriers. We describe the self-supervised world-model objective behind the programme (residual multi-horizon latent prediction, provable anti-collapse via sliced Epps–Pulley regularisation, inverse dynamics and calibrated imagination of unobserved areas) and its current reference implementation, *v8F-mini*. v8F-mini makes the hierarchy *typed*: route checkpoints, a 60-dimensional tactical code of goals, manoeuvres and continuous constraints, and a trajectory that is an exact residual on a frozen learned kinematic prior. With only 7.2 M trainable parameters, trained in four hours on one 8 GB consumer GPU, it reduces 6 s displacement error against its learned kinematic floor by 0.31/0.43 m at two training seeds in open loop. A zero-training post-map makes it better than that floor on *every* row of a pre-registered metric set. Driving on its own executed motion, the adopted model lowers 6 s error by 1.26/1.19 m and improves at least one row in each of the four metric families: longitudinal, lateral, tactical and strategic. All of it rests on an evaluation doctrine built for decisions rather than leaderboards: pre-registered bars with both outcomes committed, episode-cluster bootstrap intervals, explicit seed variance and four metric families on every evaluation. We close with the remaining frontier (assertive launch from standstill), the re-rendered simulation tier in preparation to address it, and a research roadmap toward a 300 M-parameter 4B driver deployable within 100 ms on an edge computer.

1 Introduction

Driving is a hierarchy of decisions on separated time scales. A route is chosen over tens of seconds, a manoeuvre with its parameters over seconds, a control profile over tenths of a second. Monolithic end-to-end planners collapse this structure into one mapping from pixels to waypoints. That buys simplicity and gives up four things a production driver needs:

- **data efficiency**, because every situation must be learned jointly at every scale;
- **inspectability**, because no intermediate decision is exposed;
- **self-knowledge**, because nothing in the model says when it is outside its competence;
- **efficient inference**, because the full network runs at the control rate.

TanitAD is a research programme built on the opposite bet. A driver should *imagine* the consequences of its options in a learned latent world model, at the level of abstraction where each decision lives, and expose each level's decision as a typed, measurable interface. We call the architecture *4B*, for its four brains:

- an **operative** brain for continuous control;
- a **tactical** brain for manoeuvre selection;
- a **strategic** brain for routing;
- a **fallback** brain that monitors the others and owns the minimal-risk manoeuvre.

Contributions.

(i) **The 4B formalism** (Section 3): per-level predictors in shrinking state spaces at slowing clocks, downward goal conditioning, and a gradient-isolation matrix that is *checked* rather than assumed.

(ii) **v8F-mini** (Section 4), a typed three-level instantiation over frozen foundation features:
- a tactical code with 22 goals, 12 manoeuvre classes and 26 continuous constraints;
- a trajectory parameterisation in which a zero residual reproduces a learned kinematic prior *exactly*, so every learned metre is measured against the floor.

(iii) **Mathematical analysis** (Section 5): a Lipschitz decomposition of plan error into an oracle term and propagated interface errors, the convex response of the operative layer to tactical accuracy, why whole-code teacher forcing leaves partial codes unsupported, and a three-variance account of evaluation intervals.

(iv) **Results under a decision-grade doctrine** (Sections 6 and 7): improvements over the kinematic floor in open loop and on the model's own executed motion, with at least one improved row in each of the four metric families.

(v) **A roadmap** (Sections 8 and 9): re-rendered simulation on neural reconstructions, perception adaptation, and

scaling toward the 300 M-parameter 4B budget.

2 Related work

Hierarchical latent world models.

- H-JEPA [1] stacks action-conditioned joint-embedding predictive models that predict further ahead in coarser latents, with top-down sub-goal planning.
- Hierarchical planning with latent world models [2] reports large gains of two-level sub-goals over flat planners at matched capacity.
- In driving, Orbis 2 [3], Drive-HWM [4] and S2Planner [5] apply hierarchical or multi-scale world models on NAVSIM [23].

TanitAD differs in making the hierarchy a *supervised typed interface* whose usage is measured by intervention.

Self-supervised representations.

- DINOv3 [6] supplies frozen visual features.
- V-JEPA 2 [7] and LeJEPA [8] motivate latent prediction with isotropic-Gaussian regularisation.
- LoRA [9] is the adaptation route of our perception outlook.

Why modularity can help. Shalev-Shwartz and Shashua [10] show a sample-complexity separation in favour of semantic-abstraction training, under an assumption of approximately independent module failures, which our interface analysis addresses directly.

Exposure bias and on-policy data. Scheduled sampling [11], DAgger [12], DART [13], causal confusion [14] and the limits of behaviour cloning [15] frame the mismatch between a consumer trained on ground truth and one fed its predecessor's output.

End-to-end planners.

- Representative designs: conditional imitation [16], ChauffeurNet [17], TransFuser [18], VAD [19], Hydra-MDP [20], DiffusionDrive [21] and residual planners [22].
- They are evaluated on nuPlan [24] and NAVSIM [23].

3 The 4B architecture

3.1 Levels, clocks and predictors

Let $\ell \in \{O, T, S\}$ index the operative, tactical and strategic levels. Each level owns:

- a state $z_\ell \in \mathbb{R}^{d_\ell}$;
- a clock τ_ℓ and a horizon;
- an action space a_ℓ;
- a predictor P_ℓ in its own space.

The operative state is a spatial-grid readout of the encoder, $z_O = r(f_\theta(x))$ (never a global pool). Upper states are adapters over *stopped* lower latents, $z_T = \phi_T(\mathrm{sg}[z_O])$ and $z_S = \phi_S(\mathrm{sg}[z_T])$. The dynamics are conditioned strictly downward:

$$\begin{aligned} \hat{z}_S(t{+}K) &= P_S\big(z_S(t), a_S(t)\big), \\ \hat{z}_T(t{+}k) &= P_T\big(z_T(t), a_T(t) \,\big|\, g_{\mathrm{str}}\big), \\ \hat{z}_O(t{+}j) &= P_O\big(z_O(t), (a,\kappa)_t \,\big|\, g_{\mathrm{tac}}\big). \end{aligned} \tag{1}$$

Two structural properties follow by construction and are then verified.

(1) Abstraction by down-sampling under a shrinking state. With $d_S < d_T < d_O$ and $\tau_S > \tau_T > \tau_O$, an upper level cannot represent lower-level detail even if its objective would reward it. A 15 s strategic roll is three steps of a 256-dimensional model, so only the operative path runs at the full rate.

(2) The gradient-isolation matrix. Let $G[u, v] = 1$ iff a gradient path from module u into module v is permitted. With $\mathcal{P}$ the planner and goal heads and $\mathcal{E}$ the encoder,

$$G[\mathcal{P}, \mathcal{E}] = 0, \quad G[\ell', \ell] = 0 \ \ (\ell' \text{ above } \ell), \quad G[\ell, \ell] = 1. \tag{2}$$

Every build checks it by asserting zero gradient on forbidden paths, with a deliberate-regression arm that must fail.

3.2 Self-supervised world-model objective

The programme's world model is trained without perception labels, without HD maps and without reward. Its objective is

$$\begin{aligned} \mathcal{L} = \mathcal{L}_{\mathrm{pred}} + \lambda_{\mathrm{tac}}\mathcal{L}_{\mathrm{tac}} + \lambda_{\mathrm{sig}}\big(\mathcal{S}(Z) + \mathcal{S}(\hat{Z})\big) \\ + \lambda_{\mathrm{inv}}\mathcal{L}_{\mathrm{inv}} + \lambda_{\mathrm{im}}\mathcal{L}_{\mathrm{imag}}. \end{aligned} \tag{3}$$

Residual multi-horizon prediction. $\hat{z}_{t+k} = z_t + \Delta_k(z_{t-W+1:t}, a_{t-W+1:t})$, scored by a *change-weighted* error with per-dimension weights $w \propto |z_{t+k} - z_{t+k-1}|$ normalised to unit mean. On driving data only 5–11 % of a frame changes per step, so the weighting puts the gradient where the consequences are.

Provable anti-collapse. $\mathcal{S}(\cdot)$ is the sliced Epps–Pulley statistic of LeJEPA [8]. Embeddings are projected onto M fresh unit directions, and each projection Y (a sample of size n) is tested against $\mathcal{N}(0, 1)$:

$$T_{n,\beta} = \frac{1}{n}\sum_{j,k} e^{-\frac{\beta^2(Y_j - Y_k)^2}{2}} - \frac{2}{\sqrt{1+\beta^2}}\sum_j e^{-\frac{\beta^2 Y_j^2}{2(1+\beta^2)}} + \frac{n}{\sqrt{1+2\beta^2}}. \tag{4}$$

By the Cramér–Wold theorem, normal one-dimensional projections in all directions imply a joint isotropic Gaussian, the embedding law LeJEPA shows to be optimal, and the gradients stay uniformly bounded.

Inverse dynamics. $\mathcal{L}_{\mathrm{inv}} = \|g(z_t, z_{t+1}) - a_t\|^2$ uses proprioception as free supervision and forces controllable state into the latent.

Imagination of unobserved areas. Masked sectors evolve by semi-Lagrangian advection under a learned flow, $\hat{z}_{t+1}(u) = z_t(u - v_\theta(u)\Delta t)$, refined by attention, with a per-cell log-variance $s(u)$:

$$\mathcal{L}_{\mathrm{imag}} = \sum_u w(u)\left[e^{-s(u)}\,\frac{\|\hat{z}(u) - z(u)\|^2}{2} + \frac{s(u)}{2}\right], \tag{5}$$

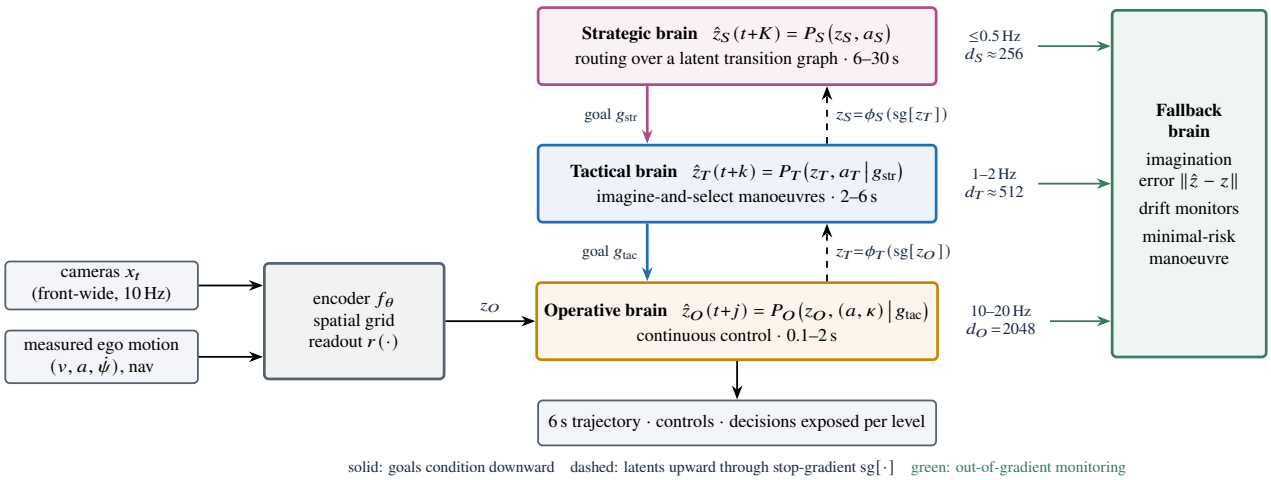


Figure 1. The 4B architecture. Four brains at separated clocks and horizons. Each predictive level owns its own predictor in its own latent space, and the state shrinks while the clock slows going up ($d_S < d_T < d_O$, $\tau_S > \tau_T > \tau_O$), so upper levels are forced to abstract. Goals condition strictly downward. Latents pass upward only through stop-gradient adapters, and the gradient-isolation matrix of Eq. (2) is tested on every build. The fallback brain consumes every level's imagination error and owns the minimal-risk manoeuvre.

with $w = 1$ on hidden cells and 0.1 on visible ones. The model must know *where it cannot know*. The variance field is the world-model-native trigger for sensor steering and for the fallback brain.

3.3 Why a hierarchy is data-efficient

Two arguments support the factorisation [2, 10].

(i) **Horizon.** Planning regret in learned models grows with the horizon, so splitting one long horizon into per-level short horizons gives a smaller bound at each level.

(ii) **Recombination.** If a driving situation is a combination of a place $p \in \mathcal{P}$, a manoeuvre $m \in \mathcal{M}$ and a control profile, a flat learner must cover $|\mathcal{P}| \cdot |\mathcal{M}|$ combinations. A hierarchy that learns each level separately covers roughly $|\mathcal{P}| + |\mathcal{M}|$.

3.4 Fallback and self-knowledge

The relative imagination error $\|\hat{z}_t - z_t\|$ is a free, always-on familiarity signal. Layered monitors arm a deterministic minimal-risk manoeuvre:

- imagination error at 10 Hz;
- routing statistics at 1 Hz;
- Mahalanobis drift at 0.1 Hz.

Their logs are the substrate for the ISMR/DSSAD incident-reporting requirements: self-monitoring *is* the evidence trail.

The programme's 263 M-parameter 4B flagship was its first arm to beat the kinematic floor in open loop: ADE@2 s 0.452 m against 0.500 m for the best of three kinematic models and 0.523 m for CTRV. On windows where kinematics diverge from the realised future, it beat CTRV by 0.80 m, and the whole of that advantage came from vision.

4 v8F-mini: a typed reference implementation

4.1 Inputs and notation

Time runs on a 0.2 s grid. For a window ending at t, the model reads:

- **scene tokens** $z_{t'} \in \mathbb{R}^{40\times1024}$, frozen DINOv3-L/16 features of the cylindrical front-wide frame, pooled to a 4×10 grid, four frames per layer;
- **ego features** $f_t = (v, a, \kappa, v^2, va, v\kappa, v^2\kappa, a\kappa)$, with curvature κ from the *measured yaw rate* (never a steering angle);
- **a navigation command** n_t;
- **a speed bucket** ℓ;
- **measured odometry** between frames.

The output is a 6 s plan $\tau \in \mathbb{R}^{60\times2}$ in the ego frame, scored on its 30 knots at 0.2 s.

4.2 The S/T/O decomposition

S ticks at about 1 Hz, T at about 2 Hz and O at 10 Hz. On the cache grid the model uses sample-and-hold with quantised ages:

- T last ticked $a_T \in \{0, 1, 2\}$ steps ago;
- S ticked $u \le 4$ steps before T, so $a_S = a_T + u$.

With $t_T = t - a_T$ and $t_S = t - a_S$,

$$\begin{aligned} \hat{k} &= f_S(z_{t_S}, f_{t_S}, n_{t_S}, \ell), \\ \hat{c} &= f_T(z_{t_T}, f_{t_T}, n_{t_T}, \ell, \Pi_{t_S\to t_T}\hat{k}), \\ \hat{\tau} &= f_O(z_t, f_t, n_t, \ell, \Pi_{t_S\to t}\hat{k}, \hat{c}, (a_T, a_S)), \end{aligned} \tag{6}$$

where $\Pi_{a\to b}$ re-expresses held checkpoints from frame a to frame b by odometry. The conditioning sets are strictly nested, $\mathrm{in}(S) \subset \mathrm{in}(T) \subset \mathrm{in}(O)$. Information flows between levels only through *held outputs*, never through shared hidden state. Each f_ℓ is a *Reader*: four learned queries cross-attend to the conditioning and scene tokens through a pre-norm transformer decoder, followed by an MLP head.

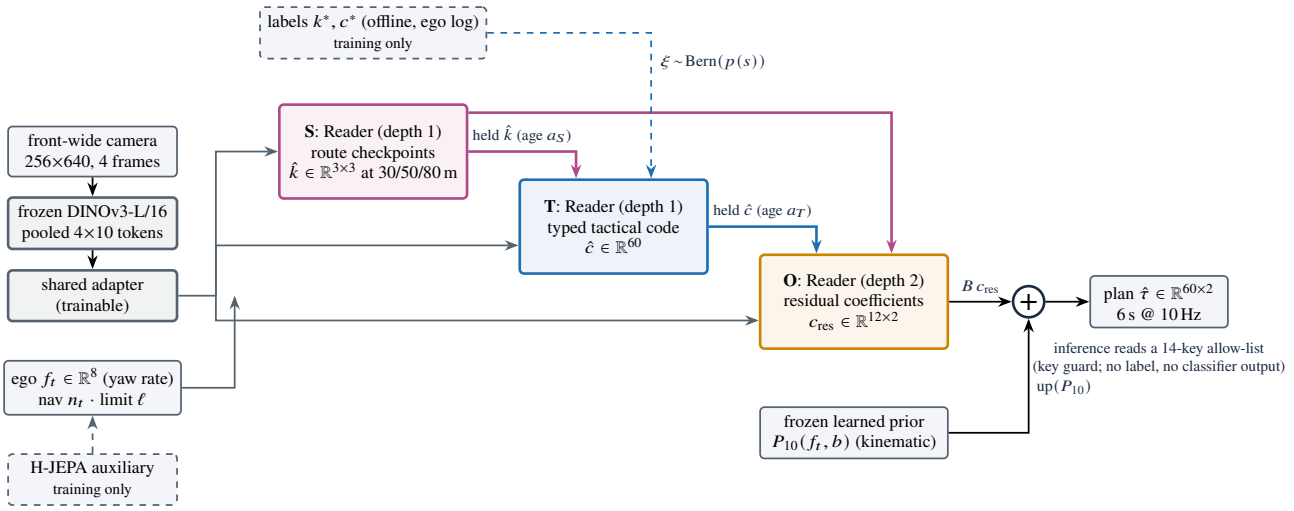


Figure 2. v8F-mini. Three learned readers over frozen DINOv3 features and a shared trainable adapter. S emits route checkpoints, T a typed 60-dimensional tactical code, and O the coefficients of a ramped cosine basis added to a *frozen* learned kinematic prior. Upper outputs are *held* with quantised ages and re-expressed in the current frame by odometry. Labels enter only through scheduled teacher forcing in training. Inference reads a fixed allow-list of inputs, enforced by a key guard.

4.3 The tactical code as a typed interface

$$c = \big[\ \underbrace{\sigma(g)}_{22\text{ goals}}\ \big|\ \underbrace{\mathrm{softmax}(\lambda_{\mathrm{lat}})}_{5}\ \big|\ \underbrace{\mathrm{softmax}(\lambda_{\mathrm{lon}})}_{7}\ \big|\ \underbrace{(\gamma-\mu)/\sigma_{\mathrm{cons}}}_{26\text{ constraints}}\ \big] \in \mathbb{R}^{60}. \tag{7}$$

The 26 constraint cells cover turn timing and geometry, longitudinal target speeds and reach times, the stop point, the lead gap and time gap, and the speed band.

An information-theoretic reading. Ground-truth codes $C^* = g(Y)$ are deterministic functions of the human future Y, so the data-processing inequality gives $I(C;Y) \geq I(C;C^*)$. Supervised agreement with C^* is therefore a lower-bound proxy for informativeness about the future. O also reads the scene, so the code is a *typed side channel*, not a strict bottleneck, and its usage is measurable by intervention (Section 7).

4.4 Trajectory as an exact residual on a kinematic prior

Let $P = P_{10}(f_t, b) \in \mathbb{R}^{30\times2}$ be a *frozen* learned prior: a damped base path re-timed by constant acceleration, plus an eight-feature MLP fitted once. Let up interpolate it to 0.1 s. The plan is

$$\tau = \mathrm{up}\big(P_{10}(f_t, b)\big) + B\,c_{\mathrm{res}}, \quad B_{ik} = \tfrac{i+1}{N}\cos\Big(\tfrac{k\pi(i+\frac{1}{2})}{N}\Big), \tag{8}$$

with $N = 60$, $k = 0, \ldots, 11$, $c_{\mathrm{res}} = s \cdot \mathrm{MLP}(h_O)$, $s = 2\,\mathrm{m}$, and the last layer zero-initialised.

Proposition 1 (Floor identity, ramp and band limit)**.**

(a) ***Floor identity.*** *$c_{\mathrm{res}} = 0$ implies $\tau = \mathrm{up}(P_{10})$ exactly, so the untrained model* is *the prior.*

(b) ***Ramp.*** *$|B_{ik}| \leq (i+1)/N$, so the residual vanishes linearly as $t \to 0$ and the plan starts at the prior's origin.*

(c) ***Band limit.*** *With $k \leq 11$ the residual contains at most eleven half-cycles over 6 s, below 1 Hz: a smooth adjustment, while the prior carries the kinematics.*

Every learned metre is thus measured against the floor it must beat. Each property is pinned by a unit test with a mutation check.

4.5 Losses and scheduled teacher forcing

The training loss is

$$\mathcal{L} = w_{\mathrm{ck}}\mathcal{L}_{\mathrm{ck}} + w_{\mathrm{tac}}\big(\mathcal{L}_{\mathrm{goal}} + \tfrac{1}{2}(\mathcal{L}_{\mathrm{lat}} + \mathcal{L}_{\mathrm{lon}}) + w_{\mathrm{cons}}\mathcal{L}_{\mathrm{cons}}\big) + w_{\mathrm{traj}}\big(\tfrac{1}{3}\mathcal{L}_{\mathrm{ADE}} + \tfrac{1}{12}\mathcal{L}_{\mathrm{FDE}}\big) + w_{\mathrm{pen}}\textstyle\sum_r \rho_r + w_{\mathrm{wm}}\mathcal{L}_{\mathrm{wm}}. \tag{9}$$

Its terms:

- **partial-label cross-entropies** $-\log \sum_{c\in\mathcal{A}} p_c$ handle the sets of allowed manoeuvres;
- **constraint cells** use a masked Huber loss;
- **rule penalties** ρ_r (speed, stop, turn, lead) are differentiable hinges that are positive *iff* a frozen rule reports a violation;
- **the world-model auxiliary** $\mathcal{L}_{\mathrm{wm}}$ is an H-JEPA objective on the shared adapter [1].

During training, O and T read a mixture of predicted and ground-truth upper outputs,

$$\tilde{c} = \xi_c\, c^* + (1-\xi_c)\,\hat{c}, \qquad \xi_c \sim \mathrm{Bernoulli}\big(p(s)\big), \quad p(s) = 1 - 0.8\min(s/S, 1), \tag{10}$$

which anneals the share of ground truth from 1.0 to 0.2 [11]. At inference $\xi \equiv 0$, and labels are never read.

4.6 Admissibility by construction

- Inference reads a fixed allow-list of 14 keys. A key guard records every access during the forward pass, and a test proves that a deliberately peeking model is caught.
- The goal path may not contain the output of any situation classifier.
- Labels may use the ego future, but inference may not.

5 Mathematical background

5.1 Plug-in composition and propagated interface error

By the chain rule, $p(y \mid x) = \sum_{k,c} p(k \mid x)\, p(c \mid x, k)\, p(y \mid x, k, c)$. Factorisation alone is an identity; its value lies in *supervision structure*. Each factor is fitted against its own offline target, and inference is a plug-in composition $\hat{y} = \mu_O(x, \hat{k}, \hat{c})$.

Proposition 2 (Error decomposition)**.** *If μ_O is L_k- and L_c-Lipschitz in its code arguments, then*

$$\|\hat{y} - y\| \le \underbrace{\|\mu_O(x, k^*, c^*) - y\|}_{\text{oracle term}} + L_c\|\hat{c} - c^*\| + L_k\|\hat{k} - k^*\|. \tag{11}$$

The oracle term is measurable by substituting ground-truth codes. The gap between the model's own code and the oracle is the propagated interface error, and it is the headroom the hierarchy offers (Section 7).

5.2 Convex response to tactical accuracy

Define the partial substitution $\hat{c}_\alpha = \hat{c} + \alpha(c^* - \hat{c})$ on the 26 constraint cells, and the recovered fraction

$$r(\alpha) = \frac{\mathrm{ADE}(\hat{c}) - \mathrm{ADE}(\hat{c}_\alpha)}{\mathrm{ADE}(\hat{c}) - \mathrm{ADE}(\hat{c}_1)}. \tag{12}$$

Linear response gives $r(\alpha) = \alpha$. We measure $r(0.25) = 0.08$–0.16, $r(0.5) = 0.20$–0.36 and $r(0.75) = 0.48$–0.73. The response is *convex*: O is flat near its own code and steep near the truth. The value of tactical accuracy therefore compounds, and the last metres of tactical precision buy the most driving.

5.3 Interface support and exposure bias

Whole-code teacher forcing trains O on the two-point mixture $p\,\delta_{c^*(x)} + (1-p)\,\delta_{\hat{c}(x)}$. A partial code $c_u = (c^*_u, \hat{c}_{\bar{u}})$ lies off that support, so its response is unconstrained by the training loss. The remedy is to widen the support: draw an independent ξ_u per unit of the code at the same marginal $p(s)$,

$$\tilde{c}_u = \xi_u c^*_u + (1 - \xi_u)\hat{c}_u, \qquad \xi_u \overset{\text{iid}}{\sim} \mathrm{Bern}\big(p(s)\big). \tag{13}$$

This keeps each cell's marginal unchanged while covering 2^7 corners per window instead of 2.

The same logic applies along time. A policy trained only on expert states incurs compounding error of order εT^2, while on-policy aggregation (DAgger [12]) reduces it to order εT. v8F-mini realises this by replacing 10 % of the training samples with the model's *own rolled-out states*, labelled from the human future.

5.4 Estimator theory: three variances, one interval

Windows are grouped into $M = 141$ episodes. For arms a, b the point estimate is the full-set mean difference $\hat{\Delta} = \frac{1}{N}\sum_w [d_a(w) - d_b(w)]$. The paired episode-cluster bootstrap resamples episodes with replacement, using the same draw for both arms, and reports percentile intervals. By the law of total variance, with training randomness θ, inference randomness ω and the episode draw E,

$$\mathrm{Var}(\hat{\Delta}) = \underbrace{\mathrm{Var}_\theta\, \mathbb{E}[\hat{\Delta} \mid \theta]}_{\text{training seeds}} + \underbrace{\mathbb{E}_\theta \mathrm{Var}_\omega\, \mathbb{E}_E[\hat{\Delta} \mid \theta, \omega]}_{\text{inference}} + \underbrace{\mathbb{E}_{\theta,\omega} \mathrm{Var}_E[\hat{\Delta} \mid \theta, \omega]}_{\text{episodes}}. \tag{14}$$

The bootstrap estimates the episode term only. TanitAD therefore reports the seed term separately, through two training seeds and a pre-registered floor $F = |\Delta_{s_0} - \Delta_{s_1}|$ with bar $2F$. For a deterministic forward pass the inference term is exactly zero, and the programme verifies that on re-evaluation (maximum per-window deviation 0.0 m).

6 Evaluation doctrine

Tiers. Every number carries its tier.

- T0: teacher-forced or oracle-conditioned. A diagnostic, never driving performance.
- Open loop: the model's own held outputs, with logged ego data.
- T1-kin: the model's *own executed motion* rebuilds the ego state at every 2 s replan (three replans of 10 executed knots per window) while the cameras stay logged.
- T2: cameras re-rendered at the car's own pose and time. In preparation, Section 8.

Four metric families, never one score. Every evaluation reports:

- *longitudinal*: speed, along-track, acceleration and distance keeping;
- *lateral*: heading, yaw rate, curvature and cross-track;
- *tactical*: manoeuvre-decision agreement;
- *strategic*: route-checkpoint error.

The families are reported per family and never pooled.

Pre-registration. Each arm has a card committed before its data. The card fixes the primary bar, the mechanism read, the refusals and *both* outcomes with their next steps. Every training run is bound by a launch gate to its commit, argv and data hashes, and every check carries a deliberate-regression arm that must fail.

Data. Training uses 4,512 PhysicalAI-AV clips (306,566 windows). Evaluation uses 141 held-out episodes (2,399 windows), disjoint by clip and by recording session.

7 Results

The hierarchy beats its floor. Against the learned kinematic floor P10, which is exactly the model's own output at a zero residual by Eq. (8), v8F-mini (M2b) lowers 6 s error by 0.312/0.432 m at two seeds, beyond a pre-registered bar of twice the seed floor (Fig. 3, Table 1). A zero-training post-map (HACC) blends O's plan with its own prior through

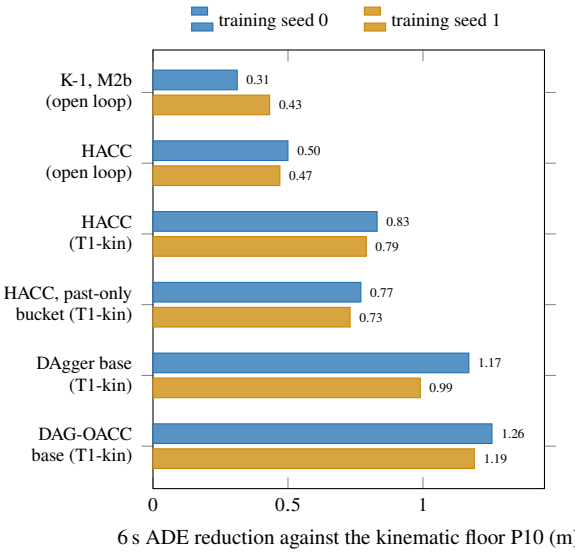


Figure 3. Six-second error reduction against the learned kinematic floor, at two training seeds. Every bar is a paired episode-cluster bootstrap contrast on 141 held-out episodes, and every one is separated from zero. The tier is given per bar.

per-step weights fitted on training windows with an acceleration term. With it, the model is separated-better than P10 on every row of a pre-registered set, including 2 s error (−0.011/−0.008 m), 6 s error (−0.50/−0.47 m) and acceleration (−0.022/−0.040 m/s^2), and separated-worse on none.

The margin survives the model's own motion. On T1-kin, where the ego state is rebuilt from the model's own executed motion at each replan, the HACC model beats P10 by 0.83/0.79 m. Two stress versions keep most of the margin:

- a speed bucket rebuilt from the last 4 s of measured speed alone keeps 0.77/0.73 m;
- freezing the camera at the first frame for 6 s as well keeps 0.73/0.64 m.

The margin therefore rests on the model's reading of the scene and of its own state, not on future information.

On-policy data makes the tactical brain robust. Replacing 10 % of training samples by the model's own rolled-out states (DAgger, $p = 0.1$) raises the tactical brain's lateral macro-F1 by 0.108/0.118 and its longitudinal macro-F1 by 0.055/0.081 on the model's own motion. It also lowers executed 6 s error by 0.54/0.40 m. It is the first training lever to clear a T1 bar on both heads. The adopted base built on it (DAG-OACC) lowers 6 s error against P10 by 1.26/1.19 m on its own motion. At least one row improves in each of the four families (longitudinal, lateral, tactical and strategic).

Every level contributes. The strategic brain's route checkpoints beat a kinematic arc at every horizon, by up to 8.8 m at 80 m. The tactical interface is used, as substituting the ground-truth code cuts 6 s error from about 2.1 to about 1.0 m. Proposition 2 reads this as roughly 1.1 m of headroom available through better tactical predictions. Because the response is convex, that headroom concentrates in the final gains in tactical precision.

Table 1. Headline results (MEASURED; paired episode-cluster bootstrap, $B = 2000$, 141 episodes; values at seed 0 / seed 1).

Result	Tier
6 s ADE −0.312 / −0.432 m vs P10, beyond the pre-registered $2F = 0.2415$ m (12.5–17.4 % lower error)	open loop
HACC post-map: better than P10 on *every* row of the pre-registered set (ADE2, ADE6, accel., curvature, speed, heading, yaw rate, decisions, consistency)	open loop
6 s ADE −0.83 / −0.79 m vs P10 on own executed motion; −0.77 / −0.73 m with a past-only speed bucket	T1-kin
Route checkpoints beat a kinematic arc by 1.2/3.3/7.8 m and 1.3/3.5/8.8 m at 30/50/80 m	T1-kin
DAgger: tactical macro-F1 +0.108/+0.118 (lateral), +0.055/+0.081 (longitudinal); 6 s ADE −0.54/−0.40 m vs its control	T1-kin
Adopted base: 6 s ADE −1.26 / −1.19 m vs P10; every family has a separated-better row	T1-kin
Ground-truth tactical code: 6 s ADE from ≈2.1 to ≈1.0 m (interface headroom)	T0

8 The frontier: assertive launch, and re-rendered simulation

With the feasibility post-step, the adopted composite meets every row of the 19-row pre-registered set but one: the *launch-crawl* share. These are situations where the human pulls away from near-standstill above 5.2 m/s and the model starts more cautiously. Every zero-training lever available on CPU has been measured against this row:

- decode levers;
- a level-from-image head;
- four doses of an external progress floor.

Even with the exact human speed supplied as a ceiling diagnostic, a bounded floor closes about half of the gap.

The analysis points to a property of the tier rather than of the planner. Logged cameras cannot show a lagging car its own consequences: the lead vehicle never pulls away on screen, and the scene never moves at the car's own pace.

T2 re-perception (in preparation). We are building a re-rendered evaluation tier on neural reconstructions of real drives:

- 62 scenes (42 launch, 20 guard), selected by a rule frozen *before* any log was read, and staged (35.7 GB);
- a harness that reproduces the T1-kin plans bit for bit from cached features, with an in-loop encoder that reproduces the cached tokens to 2.6 % relative L_2 (cosine 0.99965).

The pre-registered design runs two conditions with the same renderer: cameras rendered at the car's *own* pose and time, and at the *logged* pose re-indexed by the car's progress. Their contrast measures the camera-clock effect directly, free of the render domain gap, and both outcomes and their next levers are committed in advance.

Table 2. Roadmap (*outlook*: not yet implemented unless marked).

Item	What it delivers
T2 re-perception *(in preparation)*	a decision-grade read of the launch row on re-rendered scenes; the longitudinal tier for every later arm
Perception adaptation	LoRA on the visual encoder for motion cues; about 100 GPU-h per seed on an A40-class GPU (estimate)
Scale data	from 4,512 clips toward the full corpus; on-policy DAgger data from T2 rollouts
Scale size	from 7 M toward the 300 M 4B budget, with each level's predictor in its own latent
Strategic topology	routing over a latent transition graph, with map structure from simulation or an external corpus
Fallback brain	imagination-error monitoring, the minimal-risk manoeuvre and ISMR/DSSAD logging
Occlusion imagination	multi-step, calibrated belief roll-out of hidden sectors
Edge deployment	Jetson Thor, at most 100 ms end to end on 4 cameras
Benchmarks	NAVSIM navtest and pre-registered T2 reads

9 Outlook and research roadmap

The roadmap (Table 2) follows from the measurements. The architecture is validated at small scale on a consumer GPU, every level carries measurable value, and the remaining behaviour sits where logged-camera evaluation cannot see. Three investments turn the validated design into a driver:

- a **re-rendered simulation tier**, already in preparation;
- **perception adaptation** for motion cues;
- **scale in data and model size** toward the 4B budget.

Because the floor identity holds at every scale, and every lever is pre-registered with deliberate-regression controls, each investment yields a measured answer rather than an anecdote.

10 Conclusion

TanitAD-4B drives at every level of abstraction and holds every level accountable. Its reference implementation shows that a typed hierarchy over frozen foundation features clears a learned kinematic floor:

- in open loop, by 0.31–0.43 m of 6 s error;
- on every row of a pre-registered set, with a zero-training post-map;
- on its own executed motion, by more than a metre, with at least one improved row in each of the four metric families.

It does so with 7.2 M trainable parameters and four GPU-hours. The mathematics explains where the remaining headroom sits: in tactical precision, with convex returns, and in the evaluation tier. The roadmap names the investments that will convert the architecture into a deployable, self-aware driver.

Acknowledgements. Experiments, instruments and reports were produced with the TanitAD autonomous research system under the author's direction.

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